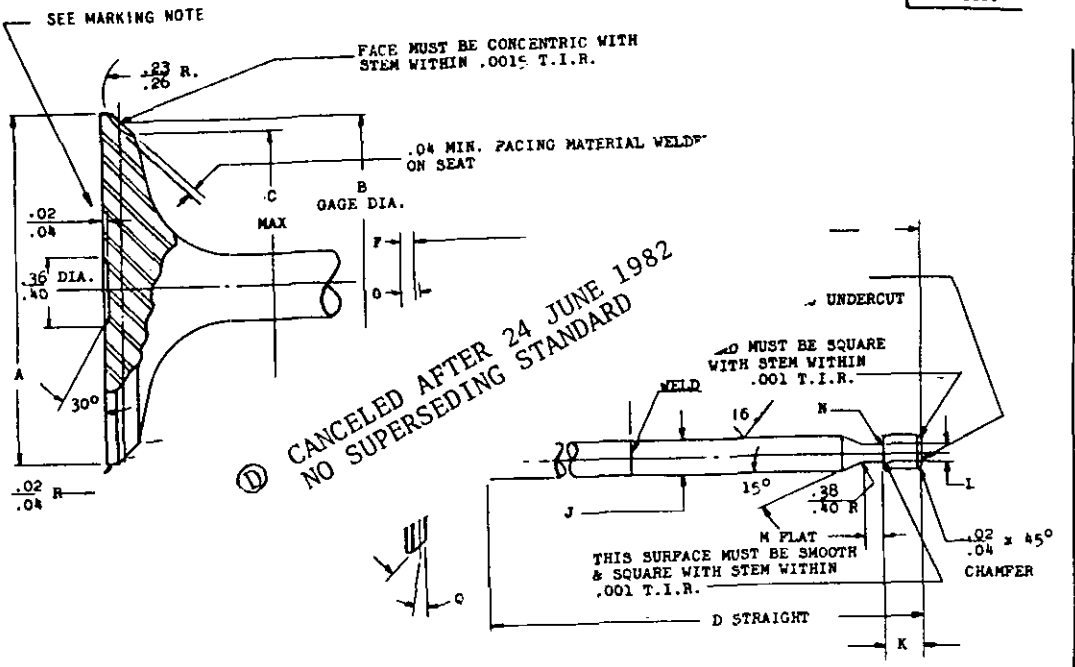
REVIEW INTEREST: ARMY - MU
AIR FORCE - 69
USER INTEREST: NAVY - YD

SUP CLASS
FED. STOCK CLASS
2805



MATERIAL : ANY COMBINATION OF VALVE HEAD AND SEAT MATERIAL SHOWN IN MATERIAL TABLE BELOW MAY BE SELECTED.

FINISH : SURFACE FINISHES TO BE IN ACCORDANCE WITH MIL-STD-10. SMOOTH GRIND JUNCTION OF STEM AND UNDER-HEAD RADIUS.

TOLERANCES: UNLESS OTHERWISE SPECIFIED. LINEAR MACHINED ± .010: ANGULAR MACHINED ±30'. DO NOT SCALE DRAWING.

DIMENSIONS: ALL DIMENSIONS ARE IN INCHES UNLESS OTHERWISE SPECIFIED.

MARKING : MS PART NUMBER AND MANUFACTURER'S IDENTIFICATION AND MATERIAL CODE TO BE MARKED AS INDICATED.

PART NUMBER: THE MS PART NUMBER CONSISTS OF THE MS NUMBER PLUS THE DASH NUMBER. EXAMPLE: MS 13999 1

TABLE I - MATERIAL BY PERCENTAGE

	STEM MATL I	HEAD MATL II	HEAD MATL III	HEAD MATL IV	FACING MATL V	SEAT MATL VI	SEAT MATL VII
CARBON	.79-.90	.15-.30	.45-.65	.15-.30	1.50 - 2.00	2.00 - 2.75	2.50 - 2.70
MANGANESE	.20 - .60	.50 - 1.50	8.00 - 10.00	5.00 - 8.00	0.30 MAX.		1.00 MAX.
PHOSPHORUS	.04 - MAX.	.04 MAX.	.04 MAX.	.04 MAX.			
SULPHUR	.04 Max.	.04 MAX.	.05 - .10	.04 MAX.			
SILICON	1.75 - 2.75	.50 - 1.50	.25 MAX.	.50 - 1.50	0.50 - 1.50	1.00 MAX.	1.5 MAX.
CHROMIUM	19.00 - 21.00	20.00 - 23.00	20.00 - 23.00	19.00 - 22.00	24.00 - 27.00	25.00 - 31.00	29.00 - 33.00
NICKEL	1.00 - 1.75	10.00 - 12.50	2.50 - 4.50	3.00 - 5.00	21.00 - 24.00	BALANCE	3.00 MAX.
NITROGEN		.30 MAX.	.25 - .45	.20 - .45			
TUNGSTEN					11.50 - 13.0	14.00 - 16.00	9.50 - 13.50
COBALT					BALANCE	9.00 - 11.00	BALANCE
IRON	BALANCE	BALANCE	BALANCE	BALANCE	2.00 MAX.	8.00 MAX.	3.00 MAX.
BORON							.05 - .20
MOLYBDENUM					0.60 MAX.		

Ⓒ ENTIRE STANDARD REVISED

APPROVED 19 June 1952 REVISED A 27 November 1953 B 24 October 1955 C 7 May 1965 D 24 JUNE 1982

THIS MIL. STANDARD HAS BEEN APPROVED BY THE DEPARTMENT OF DEFENSE AND IS MANDATORY FOR USE BY ALL DEPARTMENTS AND AGENCIES OF THE DEPARTMENT OF DEFENSE. SELECTION FOR ALL NEW ENGINEERING AND DESIGN APPLICATIONS AND FOR REPETITIVE USE SHALL BE MADE FROM THIS DOCUMENT.

P.A. Army - MO(ERDL) Other Cust Navy - YD Air Force - 69	TITLE VALVE, POPPET, ENGINE	MILITARY STANDARD MS 13999
PROCUREMENT SPECIFICATION MIL-E-11275	SUPERSEDES	SHEET 1 OF 2

DD FORM 672-1 (Coordinated) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE

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